

Sinothando Mafu

22 September 2026

SQL Fundamentals

SELECT & Filtering

Exercise 1

1. SELECT *

FROM employees;

► TABLE employees

R:	id	first_name	last_name	department	salary	hire_date	city
	1	Alice	Green	IT	70000	2020-01-10	Johannesburg
	2	Brian	Lee	HR	45000	2019-03-22	Cape Town
	3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
	4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
	5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. SELECT DISTINCT department

FROM employees;

R: ► TABLE employees

department
IT
HR
Finance
Marketing
IT

3. SELECT first_name,

last_name

FROM employees

ORDER BY salary DESC;

R: ▶ TABLE employees

first_name	last name	salary
Eva	Naidoo	72000
Alice	Green	70000
Cathy	Zulu	65000
David	Mokoena	50000
Brian	Lee	45000

4. SELECT id,
first_name,
last_name,
salary
FROM employees

▶ TABLE employees

id	first name	last name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

ORDER BY salary DESC
LIMIT 3;

5. SELECT id,
first_name,
last_name,
department
FROM employees

WHERE department = 'IT';

▶ TABLE employees

R:

id	first name	last name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. SELECT id,
 first_name,
 last_name,
 department,
 salary
 FROM employees
 WHERE department = 'Finance'
 AND salary > 60000;
 ▶ TABLE employees

R:

id	first name	last name	department	salary
3	Cathy	Zulu	Finance	65000

7. SELECT id,
 first_name,
 last_name,
 department
 FROM employees
 WHERE department = 'HR'
 OR department = 'Marketing';

▶ TABLE employees

id	first name	last name	department
2	Brian	Lee	HR
4	David	Mokona	Marketing

8. SELECT id,
 first_name,
 last_name,
 department
 FROM employees
 WHERE NOT department = 'IT';

► TABLE employees

R:	id	first name	last name	department
	2	Brian	Lee	HR
	3	Cathy	Zulu	Finance
	4	David	Mokoena	Marketing

9. SELECT id,
first_name,
last_name,
department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

► TABLE employees

R:	id	first name	last name	department
	1	Alice	Green	IT
	2	Brian	Lee	HR
	3	Cathy	Zulu	Finance
	5	Eva	Naidoo	IT

10. SELECT id,
first_name,
last_name,
department,
salary,
city
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';

►TABLE employees

R:	id	first name	last name	department	salary	city
	1	Alice	Green	IT	70000	Johannesburg
	6	Eva	Naidoo	IT	72000	Johannesburg